Principios de la Robótica Autónoma

Robótica Probabilística

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Filtro de Bayes: Estructura

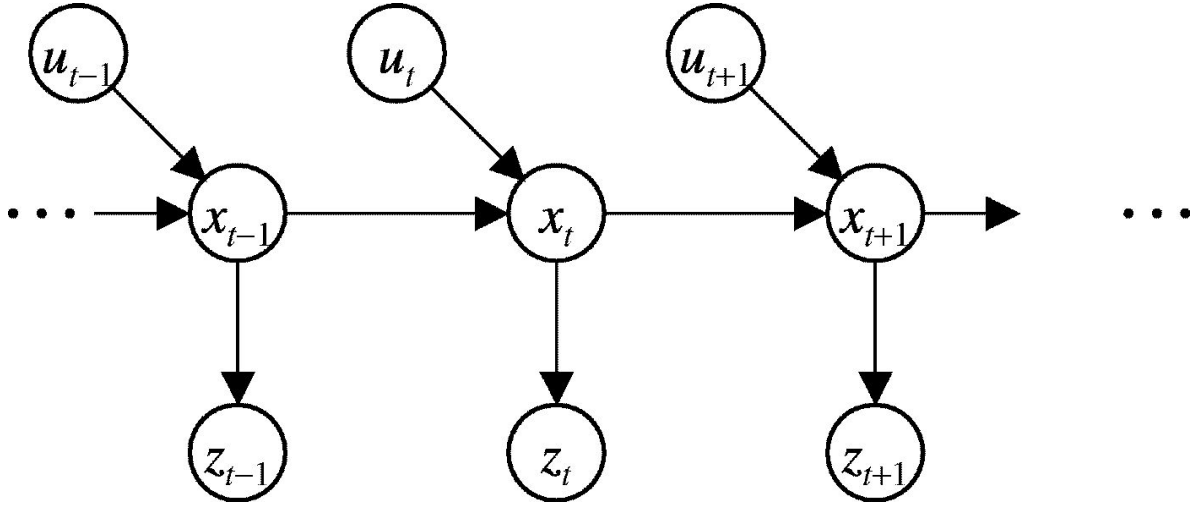
- **Dado:**
  - Una serie de observaciones *z* y acciones *u:*

$$d_t = \{u_1, z_1, \ldots, u_t, z_t\}$$

  - Modelo del Sensor $P(z \mid x)$
  - Modelo de Acción $P(x \mid u, x')$
  - Probabilidad a priori del estado del sistema $P(x)$
- **Se desea:**
  - Estimar el estado *X* de un sistema dinámico
  - La probabilidad a posteriori del estado, también llamada **Belief**:

$$Bel(x_t) = P(x_t \mid u_1, z_1, \ldots, u_t, z_t)$$

Suposición de Markov



$$P(z_t \mid x_{0:t}, z_{1:t-1}, u_{1:t}) = P(z_t \mid x_t)$$
$$P(x_t \mid x_{1:t-1}, z_{1:t-1}, u_{1:t}) = P(x_t \mid x_{t-1}, u_t)$$

Suposición subyacente

- Mundo estático
- Ruido independiente
- Modelo perfecto, sin errores de aproximación

Filtro de Bayes

z = observación
u = acción
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Bayes $$= \eta\, p\left(z_t | x_t, z_{1:t-1}, u_{1:t}\right) p\left(x_t | z_{1:t-1}, u_{1:t}\right)$$ donde η es una constante de normalización

Filtro de Bayes

z = observación
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$$
\begin{aligned}
bel(x_t) &= p\left(x_t | z_{1:t}, u_{1:t}\right) \\
&= \eta\, p\left(z_t | x_t, z_{1:t-1}, u_{1:t}\right) p\left(x_t | z_{1:t-1}, u_{1:t}\right) \\
&= \eta\, p\left(z_t | x_t\right) p\left(x_t | z_{1:t-1}, u_{1:t}\right)
\end{aligned}
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donde η es una constante de normalización

Markov

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\end{aligned}
$$

donde η es una constante de normalización

Total prob.

Filtro de Bayes

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donde η es una constante de normalización

Indep.

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&= \eta\, p\left(z_t | x_t\right) \int p\left(x_t | x_{t-1}, u_t\right) bel(x_{t-1}) dx_{t-1}
\end{aligned}
$$

donde η es una constante de normalización

Término recursivo!

$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Algoritmo **Filtro_de_Bayes** (*Bel(x)*, *d*):

η=0

If *d* is a perceptual data item *z* then

For all *x* do

$$Bel'(x) = P(z \mid x) Bel(x)$$

$$\eta = \eta + Bel'(x)$$

For all *x* do

$$Bel'(x) = \eta^{-1} Bel'(x)$$

Else if *d* is an action data item *u* then

For all *x* do

$$Bel'(x) = \int P(x \mid u, x') Bel(x') dx'$$

Return *Bel'(x)*

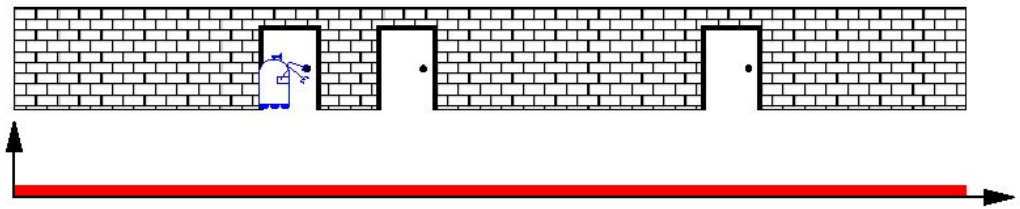
Los filtros de Bayes son muy usados !

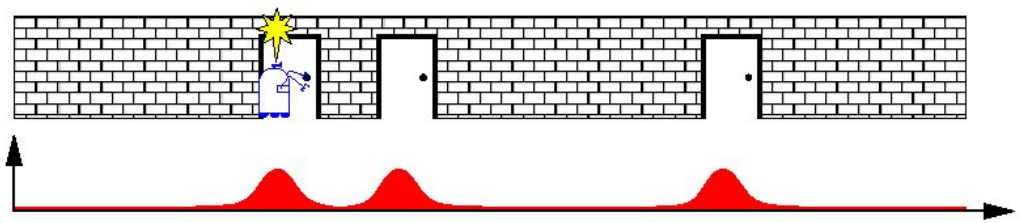
$$Bel(x_t) = \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

- Filtros de Kalman
- Filtros de Partículas
- Modelos ocultos de Markov
- Redes Bayesianas dinámicas
- Procesos de decisión de Markov parcialmente observables (POMDPs)

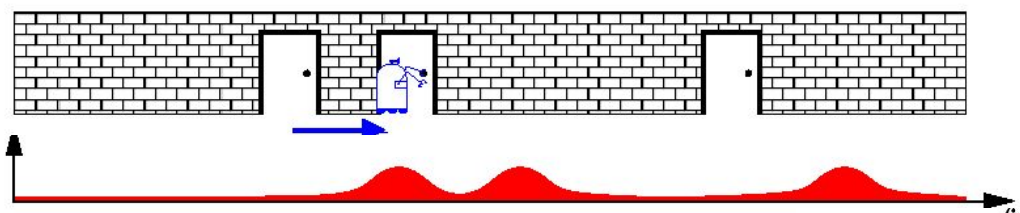
Localización Probabilística

$$bel(x_t) = \eta\, p\,(z_t|x_t) \int p\,(x_t|x_{t-1}, u_t)\, bel(x_{t-1}) dx_{t-1}$$

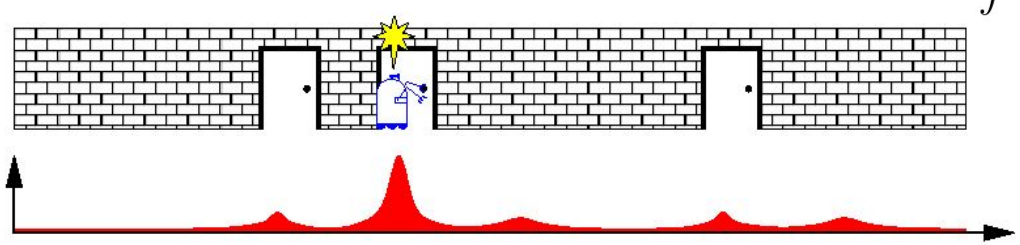




$$bel(x_t) = \eta\, p\,(z_t|x_t)\, \overline{bel}(x_t)$$



$$\overline{bel}(x_t) = \int p\,(x_t|x_{t-1}, u_t)\, bel(x_{t-1}) dx_{t-1}$$



$$bel(x_t) = \eta\, p\,(z_t|x_t)\, \overline{bel}(x_t)$$

Resumen

- La regla de Bayes nos permite calcular probabilidades que de otra forma serían muy complejas.

- Con la suposición de Markov, la actualización recursiva Bayesiana es una forma eficiente de combinar evidencia.

- Los filtros de Bayes son una herramienta probabilística para estimar el estado de un sistema dinámico.